

(3) 重积分与累次积分相等:

$$\int_Z f(z) \lambda(dz) = \int_{X \times Y} f(x, y) \mu(dx) \nu(dy) \quad (3.3.22)$$

$$= \int_X \mu(dx) \int_Y f(x, y) \nu(dy) \quad (3.3.23)$$

$$= \int_Y \nu(dy) \int_X f(x, y) \mu(dx). \quad (3.3.24)$$

习 题

1. 求积分:

$$\int_0^\infty (e^{-ax^2} - e^{-bx^2}) \frac{1}{x} dx, \quad 0 < a < b.$$

2. 求积分:

$$\int_0^\infty \frac{1}{x} (e^{-ax} - e^{-bx}) \sin x dx, \quad 0 < a < b.$$

3. 设 $f \in L([0, 1] \times [0, 1])$, 证明

$$\int_0^1 dx \int_0^x f(x, y) dy = \int_0^1 dy \int_y^1 f(x, y) dx.$$

4. 设 $f \in L([0, a])$, $g(x) = \int_x^a \frac{f(t)}{t} dt$, 证明

$$\int_0^a g(x) dx = \int_0^a f(x) dx.$$

5. 设 $f \in L([a, b])$, 证明

$$\int_a^b f(x) dx \int_a^x f(y) dy = \frac{1}{2} \left[\int_a^b f(x) dx \right]^2.$$

6. 设 $f \in L([a, b])$, 证明

$$\int_a^b dx \int_a^b dy \int_x^y f(t) dt = 0.$$

7. 设 $f, g \in L([a, b])$, $F(x) = \int_a^x f(t) dt$, $G(x) = \int_a^x g(t) dt$, 证明

$$\int_a^b F(x) g(x) dx = F(x) G(x) \Big|_a^b - \int_a^b f(x) G(x) dx.$$